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Thermodynamic assessments of the Pu–Zn and La–Pu systems

X.J. Liu [a,b], Q. He [a], D. Wang [a], Y. Lu [a,b], M.H. Chen [a], J.P. Jia [c], C.P. Wang [a,b,*]

[a] Department of Materials Science and Engineering, College of Materials, Xiamen University, Xiamen 361005, PR China
[b] Research Center of Materials Design and Applications, Xiamen University, Xiamen 361005, PR China
[c] Science and Technology on Surface Physics and Chemistry Laboratory, Mianyang 621907, PR China

HIGHLIGHTS

- The Pu–Zn and La–Pu binary systems have been assessed by using the CALPHAD method for the first time.
- The intermetallic compounds which have homogeneity range were described by the two-sublattice model.
- The parameters of the Pu–Zn and La–Pu binary systems can serve as a part of the thermodynamic database for nuclear materials.
- The calculated results are in good agreement with the experimental data.

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ABSTRACT

The thermodynamic description of the Pu–Zn and La–Pu systems were obtained using the CALPHAD (Calculation of Phase Diagrams) method on the basis of experimental information for phase equilibria and thermodynamic properties. The Gibbs free energies of the liquid, (εPu), (δ′Pu), (δPu), (γLa) and (βLa) phases were described by the subregular solution model. Seven intermetallic compounds ($PuZn_2$, $Pu_{13}Zn_{58}$, Pu_3Zn_{22}, Pu_2Zn_{17}, κ, κ′, and λ) were modeled by the sublattice model. A reasonable agreement between the present calculated results and experimental data has been achieved with a set of optimized parameters.

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1. Introduction

In recent years, the price volatility of traditional fossil fuel, the problems on the nuclear power plants' reliability, and the ever-increasing concern on the environmental protection issue induced by greenhouse gas emissions have aroused the world's attention on how to use nuclear energy as a safe and effective energy source for power generation [1–4]. In order to address this problem, it is necessary to develop advanced nuclear materials possessing better performance. On account of the stringent experimental conditions, the traditional methods conducted by former researchers with many experimental trials are unpractical. A better approach is to study the relevant phase diagrams of nuclear materials that are vital for the development of advanced nuclear metallic fuels in the first place. Thermodynamic assessments in nuclear material systems have been conducted by the present authors using CALPHAD (Calculation of Phase Diagrams) method [5–10] so as to develop a thermodynamic database for nuclear materials. As a part of this database, the present work focuses on thermodynamic assessments of the Pu–Zn and La–Pu binary systems based on the CALPHAD technique from available experimental data [13,16–25]. Pu-based alloys have shown good performance as fuels for fast reactors in many experiments carried out in the United States and Europe [11]. Pu fission in nuclear power reactors can also be used to provide a large amount of electricity globally and is the basis for many nuclear weapons [12]. Additionally, Zn has been utilized as a solvent in several pyrometallurgical processes for the purification of metallic nuclear fuels [13], and the thermodynamic evaluation between rare earth element and plutonium is rare. Therefore, it would be meaningful to gain a comprehensive understanding of the phase diagrams in the Pu–Zn and La–Pu binary systems.

2. Thermodynamic models

The information of the stable solid phases and the used models in the Pu–Zn and La–Pu systems are presented in Table 1.

2.1. Solution phases

In the A–B system, corresponding to the Pu–Zn, and La–Pu systems, Gibbs free energy of the solution phases (liquid, (εPu), (δPu),

* Corresponding author at: Department of Materials Science and Engineering, College of Materials, Xiamen University, Xiamen 361005, PR China. Tel.: +86 592 2180606; fax: +86 592 2187966.
E-mail address: wangcp@xmu.edu.cn (C.P. Wang).

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Table 1
The stable solid phases and the used models in the Pu–Zn and La–Pu systems.

System	Phase	Prototype	Strucktur–bericht designation	Modeling phase	Used models
Pu–Zn	(εPu)	W	A2	(Pu, Zn)	SSM
	(δ′Pu)	In	A6	(Pu, Zn)	SSM
	(δPu)	Cu	A1	(Pu, Zn)	SSM
	(γPu)	γPu	–	(Pu)	SSM
	(βPu)	βPu	–	(Pu)	SSM
	(αPu)	αPu	–	(Pu)	SSM
	$PuZn_2$	Cu_2Mg	C15	$(Pu)_{0.3333}(Zn)_{0.6667}$	SM
	$Pu_{13}Zn_{58}$	$Gd_{13}Zn_{58}$	–	$(Pu)_{0.1831}(Pu, Zn)_{0.8169}$	SM
	Pu_3Zn_{22}	Pu_3Zn_{22}	–	$(Pu)_{0.12}(Zn)_{0.88}$	SM
	Pu_2Zn_{17}	–	–	$(Pu)_{0.1053}(Zn)_{0.8947}$	SM
	κ	–	–	$(Pu)_{0.1}(Zn)_{0.9}$	SM
	κ′	–	–	$(Pu)_{0.1}(Zn)_{0.9}$	SM
	λ	–	–	$(Pu)_{0.095}(Zn)_{0.905}$	SM
	(Zn)	Mg	A3	(Zn)	SSM
La–Pu	(γLa)	W	A2	(La, Pu)	SSM
	(βLa)	Cu	A1	(La, Pu)	SSM
	(αLa)	αLa	A3′	(La)	SSM
	(εPu)	W	A2	(La, Pu)	SSM
	(δ′Pu)	In	A6	(La, Pu)	SSM
	(δPu)	Cu	A1	(La, Pu)	SSM
	(γPu)	γPu	–	(Pu)	SSM
	(βPu)	βPu	–	(Pu)	SSM
	(αPu)	αPu	–	(Pu)	SSM

SSM: subregular solution model.
SM: sublattice model.

(δ′Pu), (γLa), (βLa)) were described by a subregular solution model. The molar Gibbs free energy of each solution phase can be expressed as:

$$G_m^{\phi} = \sum_{i=A,B} {}^0G_i^{\phi} x_i + RT \sum_{i=A,B} x_i \ln x_i + {}^EG_m^{\phi}, \quad (1)$$

where ${}^0G_i^{\phi}$ is the molar Gibbs free energy of pure component i in the respective reference state with the structure ϕ in a nonmagnetic state, which is taken from the SGTE pure element database [14]. x_i represents the mole fraction of component i; R denotes the gas constant; T is the absolute temperature in Kelvin. The excess Gibbs free energy ${}^EG_m^{\phi}$ is given by a Redlich–Kister polynomial [15] as follows:

$${}^EG_m^{\phi} = L_{A,B}^{\phi} x_A x_B, \quad (2)$$

where $L_{A,B}^{\phi}$ is the interaction energy between A and B atoms, and is expressed as:

$$\begin{aligned} L_{A,B}^{\phi} &= {}^0L_{A,B}^{\phi} + {}^1L_{A,B}^{\phi}(x_A - x_B) + {}^2L_{A,B}^{\phi}(x_A - x_B)^2 + \ldots \\ &= \sum_{m=0}^{n} {}^mL_{A,B}^{\phi}(x_A - x_B)^m, \end{aligned} \quad (3)$$

$${}^mL_{A,B}^{\phi} = a + bT, \quad (4)$$

where the parameters of a and b were optimized based on the experimental data in the present work.

2.2. Stoichiometric intermetallic compounds

In the Pu–Zn system, six intermetallic compounds ($PuZn_2$, Pu_3Zn_{22}, Pu_2Zn_{17}, κ, κ′ and λ) were treated as stoichiometric compounds, as no experimental data was available for the homogeneity range of each compound. The Gibbs free energy for per mole atoms of the formula unit $(Pu)_p(Zn)_q$ can be expressed as follows:

$$\begin{aligned} \Delta^0G_f^{Pu_pZn_q} &= G_m^{Pu_pZn_q} - p/(p+q)^0G_{Pu}^{ref} - q/(p+q)^0G_{Zn}^{ref} \\ &= a + bT + cTLn(T) + dT^2 + eT^3, \end{aligned} \quad (5)$$

where the $\Delta^0G_f^{Pu_pZn_q}$ indicates the standard Gibbs free energy of formation per mole atoms of the stoichiometric compound from the pure elements, The terms ${}^0G_{Pu}^{ref}$ and ${}^0G_{Zn}^{ref}$ represent the molar Gibbs free energy of pure element Pu and Zn with its defined reference structure in a nonmagnetic state, and the parameters of a, b, c, d and e were evaluated in the present work. The parameters of a and b are related to the enthalpies and entropies, while c, d and e are mainly related to the heat capacity of each compound.

2.3. Non-stoichiometric intermetallic compounds

The $Pu_{13}Zn_{58}$ phase was considered as a non-stoichiometric intermetallic compound. The Gibbs free energy of formation per mole of formula unit $(Pu)_{13}(Pu, Zn)_{58}$ can be expressed as the following equation with reference to the pure elements in their non-magnetic states:

$$\begin{aligned} G^{(Pu)_{13}(Pu,Zn)_{58}} &= y_{Pu}^{II}\,{}^0G_{Pu:Pu}^{(Pu)_{13}(Pu,Zn)_{58}} + y_{Zn}^{II}\,{}^0G_{Pu:Zn}^{(Pu)_{13}(Pu,Zn)_{58}} \\ &\quad + 0.8169RT(y_{Pu}^{II}\ln y_{Pu}^{II} + y_{Zn}^{II}\ln y_{Zn}^{II}) \\ &\quad + 0.8169 y_{Pu}^{II} y_{Zn}^{II} L_{Pu:Pu,Zn}, \end{aligned} \quad (6)$$

where y_{Pu}^{II} and y_{Zn}^{II} are the site fractions of component Pu and Zn located on sublattice II, and the parameter ${}^0G_{Pu:*}^{(Pu)_{13}(Pu,Zn)_{58}}$ represents the Gibbs free energy of the compound phase when the first sublattice is occupied by element Pu, while the second sublattice is fully occupied by element Pu or Zn, respectively. $L_{Pu:Pu,Zn}$ is the interaction parameter between Pu and Zn in the second sublattice, when the first sublattice is completely occupied by element Pu. The parameters of ${}^0G_{Pu:Pu}^{(Pu)_{13}(Pu,Zn)_{58}}$, ${}^0G_{Pu:Zn}^{(Pu)_{13}(Pu,Zn)_{58}}$ and $L_{Pu:Pu,Zn}$ were evaluated in the present work.

3. Experimental information

3.1. The Pu–Zn system

The Pu–Zn system is composed of two solution phases ((εPu) and (δPu)) and seven intermediate compounds ($PuZn_2$, $Pu_{13}Zn_{58}$, Pu_3Zn_{22}, Pu_2Zn_{17}, κ, κ′ and λ). The temperatures and compositions of invariant reactions in this system have been investigated by many researchers [13,16–25].

The composition region from 0 to 65 at.% Zn of the assessed Pu–Zn phase diagram was on the basis of Cramer's study [18,19]. The region of (εPu) and (δPu) was defined, and the eutectoid point of

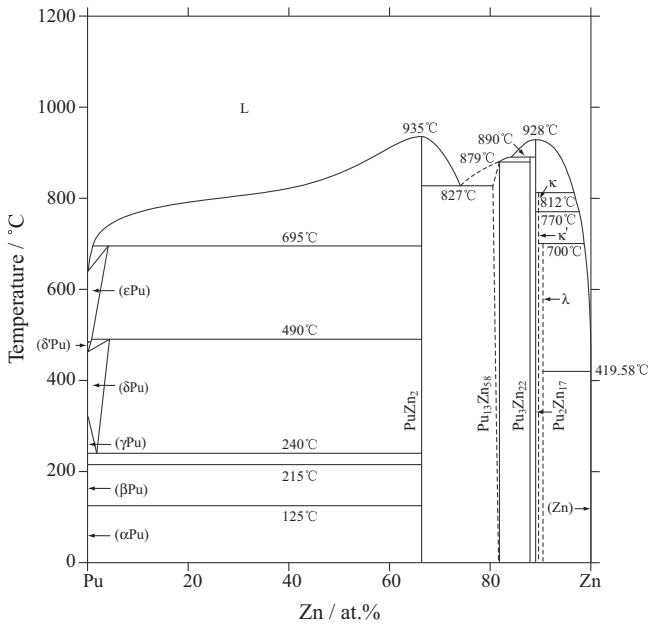

Fig. 1. The phase diagram of the Pu–Zn system reviewed by Massalski et al. [25].

the reaction, (δPu) ↔ $PuZn_2$ + (γPu), was reported to be 1.8 at.% Zn at 240 °C by Cramer [18]. The results [18] also indicated little solubility of Zn in (γPu) phase. Later on, Albrecht [20] verified a eutectic reaction at 830 °C with critical point at 74 at.% Zn by visual thermal analysis, and proved congruent temperature of $PuZn_2$ phase to be 935 ± 3 °C [20]. Due to controversial opinions on the composition of intermediate compounds, a more precise research, from 75 to 98 at.% Zn, was proceeded by Cramer and Wood [21]. Additionally, Cramer et al. [21] detected a new phase, tentatively named $PuZn_{7.7}$, which was produced by a peritectic reaction at 890 °C, and three hexagonal super-lattice phases (κ, κ′ and λ) presenting thermal effects at 811 °C, 770 °C and 700 °C, respectively. The composition of most Zn-rich phases was estimated to be close to that of $PuZn_{12}$ by Johnson et al. [13], but Cramer et al. [21] failed to confirm this information in their research. The exact compositions of these three phases are still unknown, but the relative positions on the phase diagram have been established by metallographic and thermal analyses [21]. The exact compositions of κ, κ′ and λ phases in our present work are based on the recommended values given in Massalski's work [25]. Apart from these, Pu_2Zn_{17} was identified with a rhombohedral symmetry [16], which melts congruently at 928 °C [21], and Pu_2Zn_9 phase was demonstrated by Cramer et al. [21] to be the result of a peritectic transforming, L + $PuZn_{7.7}$ ↔ Pu_2Zn_9, at 879 °C with a composition very close to the stoichiometric composition. In addition, an

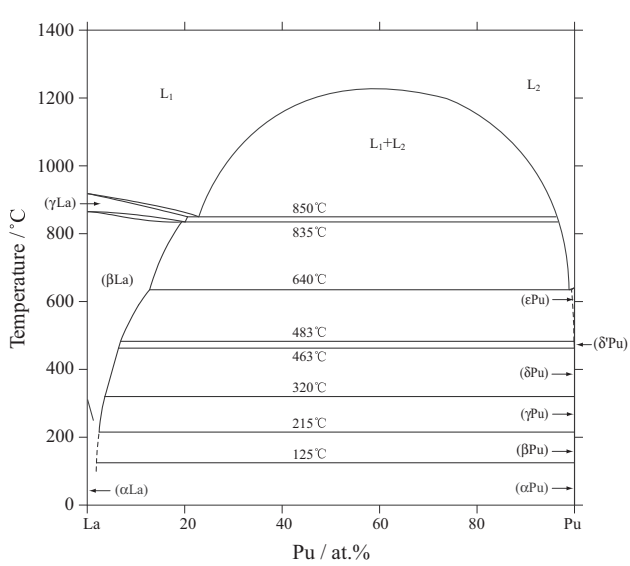

Fig. 2. The phase diagram of the La–Pu system reviewed by Okamoto et al. [31].

Table 2
The optimized thermodynamic parameters in the Pu–Zn system.

Parameters in each phase (J/mol)
Liquid phase, format (Pu, Zn)
${}^0L^{Liq}_{Pu,Zn} = -34900 + 17.6T$
${}^1L^{Liq}_{Pu,Zn} = 37600$
${}^2L^{Liq}_{Pu,Zn} = -31600 + 4.8T$
${}^3L^{Liq}_{Pu,Zn} = 19300 - 2T$
bcc (εPu) phase, format (Pu, Zn)
${}^0L^{(\varepsilon Pu)}_{Pu,Zn} = -18000 + 18.07T$
fcc (δPu) phase, format (Pu, Zn)
${}^0L^{(\delta Pu)}_{Pu,Zn} = -25400 + 25.9T$
A6 (δ′Pu) phase, format (Pu, Zn)
${}^0L^{(\delta' Pu)}_{Pu,Zn} = -4500$
$PuZn_2$ phase, format $(Pu)_{0.3333}(Zn)_{0.6667}$
$G^{PuZn2}_{Pu:Zn} = -21790 + 21.86T - 1.46TLn(T) - 0.0094T^2 + 1.21 * 10^{-6}T^3 + 0.3333{}^0G^{(\alpha Pu)}_{Pu} + 0.6667{}^0G^{(Zn)}_{Zn}$
$Pu_{13}Zn_{58}$ phase, format $(Pu)_{0.1831}(Pu, Zn)_{0.8169}$
$G^{Pu13Zn58}_{Pu:Zn} = -26287 + 18.313T - 0.65TLn(T) - 0.006T^2 + 0.35 * 10^{-6}T^3 + 0.1831{}^0G^{(\alpha Pu)}_{Pu} + 0.8169{}^0G^{(Zn)}_{Zn}$
$G^{Pu13Zn58}_{Pu:Pu} = 20500 + G^{(\alpha Pu)}_{Pu}$
${}^0L^{Pu13Zn58}_{Pu:Pu,Zn} = 4000 - 6T$
Pu_3Zn_{22} phase, format $(Pu)_{0.12}(Zn)_{0.88}$
$G^{Pu3Zn22}_{Pu:Zn} = -25795 + 8.17T + 0.12{}^0G^{(\alpha Pu)}_{Pu} + 0.88{}^0G^{(Zn)}_{Zn}$
Pu_2Zn_{17} phase, format $(Pu)_{0.1053}(Zn)_{0.8947}$
$G^{Pu2Zn17}_{Pu:Zn} = -26045 + 8.76T + 0.1053{}^0G^{(\alpha Pu)}_{Pu} + 0.8947{}^0G^{(Zn)}_{Zn}$
κ phase, format $(Pu)_{0.1}(Zn)_{0.9}$
$G^{\kappa}_{Pu:Zn} = -25044 + 8.386T + 0.1{}^0G^{(\alpha Pu)}_{Pu} + 0.9{}^0G^{(Zn)}_{Zn}$
κ′ phase, format $(Pu)_{0.1}$ $(Zn)_{0.9}$
$G^{\kappa'}_{Pu:Zn} = -25141 + 8.479T + 0.1{}^0G^{(\alpha Pu)}_{Pu} + 0.9{}^0G^{(Zn)}_{Zn}$
λ phase, format $(Pu)_{0.095}(Zn)_{0.905}$
$G^{\lambda}_{Pu:Zn} = -24007 + 8.024T + 0.095{}^0G^{(\alpha Pu)}_{Pu} + 0.905{}^0G^{(Zn)}_{Zn}$

Table 3
The optimized thermodynamic parameters in the La–Pu system.

Parameters in each phase (J/mol)
Liquid phase, format (La, Pu)
${}^0L^{Liq}_{La,Pu} = 29020 - 5T$
${}^1L^{Liq}_{La,Pu} = -7000$
${}^2L^{Liq}_{La,Pu} = 2200 - T$
fcc (γLa, εPu) phase, format (La, Pu)
${}^0L^{fcc}_{La,Pu} = 25250 - 1.9T$
${}^1L^{fcc}_{La,Pu} = -15400 + 6.3T$
${}^2L^{fcc}_{La,Pu} = 4300 + 1.05T$
bcc (βLa, δPu) phase, format (La, Pu)
${}^0L^{bcc}_{La,Pu} = 30780 - 6.8T$
${}^1L^{bcc}_{La,Pu} = -6410$
${}^2L^{bcc}_{La,Pu} = 7490 - 4.1T$
A6 (δ′Pu) phase, format (La, Pu)
$G^{(\delta' Pu)}_{La} = 5000 + G^{(\alpha La)}_{La}$
${}^0L^{(\delta' Pu)}_{La,Pu} = 52900 - 25.03T$

observable homogeneity area with an uncertain range was also identified by Cramer et al. [21]. Afterwards, it was proved that Pu_2Zn_9 may be replaced by $Pu_{13}Zn_{58}$ for its structural similarity to $Gd_{13}Zn_{58}$ [22] and the exact formula of $PuZn_{7.7}$ is Pu_3Zn_{22} [24]. The liquidus curve of the Zn-rich portion is mainly based on the work of Johnson et al. [13] and Elliott et al. [17]. According to the above experimental data, Massalski reviewed the Pu–Zn phase diagram [25], as shown in Fig. 1.

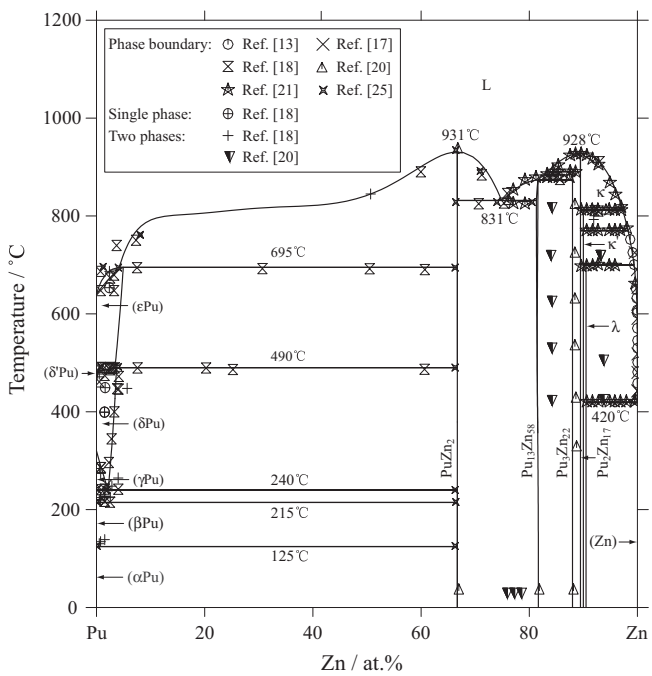

Fig. 3a. The calculated phase diagram of the Pu–Zn system compared with the experimental data [13,17,18,20,21,25].

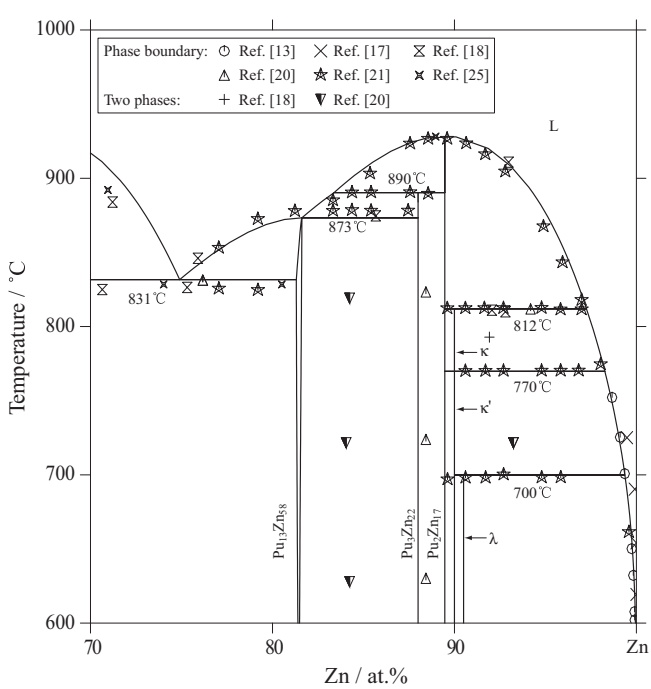

Fig. 3b. The calculated phase diagram of the Pu–Zn system in the Zn-rich portion compared with the experimental data [13,17,18,20,21,25].

In addition, the enthalpies of phase formation of the $PuZn_2$, Pu_2Zn_9, Pu_3Zn_{22}, Pu_2Zn_{17} compounds was predicted by Miedema [26]. Later on, the entropy and enthalpy of formation of Pu_2Zn_{17} in the temperature span from 25 °C to 425 °C was determined by Chiotti et al. [27] based on the EMF (electromotive forces) measurements of Johnson et al. [13]. The EMF of the galvanic cell in Pu–97.8Zn (at.%) alloy in Johnson's investigation [13] were also taken into consideration in this work.

3.2. The La–Pu system

The La–Pu phase diagram is characterized by a liquid miscibility gap, large regions of solid solubility of Pu in the (γLa) and (βLa) phases and the absence of intermediate phases. Its phase diagram has been investigated by many investigators [23,25,28–32].

The first comprehensive investigation into the phase diagram of Pu–La system was carried out by Ellinger et al. [30]. Ellinger's investigation [30] not only denoted that the monotectic reaction, $L_1 \leftrightarrow (\gamma La) + L_2$, and the catatectic reaction, $(\gamma La) \leftrightarrow L_2 + (\beta La)$, occurs at 23 at.% Pu, 850 ± 3 °C and 20 at.% Pu, 835 °C respectively, but also indicated that the solubility of plutonium in (βLa) and (γLa) was approximately 19 and 20 at.% Pu respectively by the thermal, micrographic, and X-ray diffraction techniques. These

Table 4
Experimental data and calculated invariant reactions in the Pu–Zn system [18,20,21,25].

Reaction type	Reaction	Zn (at.%)			T (°C)	Reference
Congruent	$L \leftrightarrow PuZn_2$		66.7		935	[25]
			66.7		935 ± 3	[20]
			66.7		931	This work
Congruent	$L \leftrightarrow Pu_2Zn_{17}$		89.5		928	[25]
			89.5		928	[21]
			89.5		928	This work
Polymorphic	$\kappa \leftrightarrow \kappa'$		90.0		770	[25]
			–		770 ± 3	[21]
			90.0		770	This work
Polymorphic	$(\gamma Pu) \leftrightarrow (\beta Pu)$		~0		215	[25]
			~0		215	This work
Polymorphic	$(\beta Pu) \leftrightarrow (\alpha Pu)$		–		125	[25]
			0		125	This work
Eutectic	$L \leftrightarrow PuZn_2 + Pu_{13}Zn_{58}$	74.0	66.7	–	830	[18]
		74.0	66.7	80.7	827	[25]
		74.0	66.7	–	830	[20]
		–	–	–	827 ± 5	[21]
		74.9	66.7	81.3	831	This work
Eutectic	$L \leftrightarrow \lambda + (Zn)$	100	90.5	100	420	[25]
		–	–	–	419	[20]
		–	–	–	420	[21]
		100	90.5	100	420	This work
Eutectoid	$(\delta Pu) \leftrightarrow PuZn_2 + (\gamma Pu)$	1.8	–	–	240	[18]
		1.9	66.7	0.1	240	[25]
		2.1	66.7	~0	240	This work
Peritectic	$L + Pu_2Zn_{17} \leftrightarrow Pu_3Zn_{22}$	84.0	89.5	88.0	890	[25]
		–	89.5	–	890 ± 3	[21]
		83.4	89.5	88.0	890	This work
Peritectic	$L + Pu_3Zn_{22} \leftrightarrow Pu_{13}Zn_{58}$	81.6	88.0	81.7	879	[25]
		~81.7	–	–	879 ± 3	[21]
		81.6	88.0	81.6	873	This work
Peritectic	$L + Pu_2Zn_{17} \leftrightarrow \kappa$	96.7	89.5	90	812	[25]
		–	89.5	–	812 ± 3	[21]
		97.2	89.5	90	812	This work
Peritectic	$L + \kappa' \leftrightarrow \lambda$	98.7	90	90.5	700	[25]
		–	–	–	700 ± 8	[21]
		99.4	90	90.5	700	This work
Peritectic	$L + PuZn_2 \leftrightarrow (\varepsilon Pu)$	–	–	–	695	[18]
		1.2	66.7	4.0	695	[25]
		4.2	66.7	4.9	695	This work
Peritectoid	$PuZn_2 + (\varepsilon Pu) \leftrightarrow (\delta Pu)$	–	–	–	490	[18]
		66.7	0.9	4.0	490	[25]
		66.7	3.4	3.7	490	This work
Peritectoid	$(\varepsilon Pu) + (\delta Pu) \leftrightarrow (\delta' Pu)$	0.3	0.4	0.4	483	This work

results were in line with the previous conclusions of Poole [28] with minor discrepancies. Ellinger et al. [30] found the eutectic transforming, $L_2 \leftrightarrow (\beta La) + (\varepsilon Pu)$, at 632 ± 4 °C, and confirmed Poole's speculation [28] about the existence of liquid miscibility gap in this system as well. However, no effort was made to determine the accurate solubility of the liquid miscibility gap. Much later, Okamoto [31] reviewed the Pu–La phase diagram based on previous researches, as presented in Fig. 2.

No experimental thermodynamic data is available in the La–Pu binary system, but Takeuchi et al. [33] estimated the enthalpy of mixing of liquid phase at 50 at.% Pu to be 7 kJ/mol by Miedema's model.

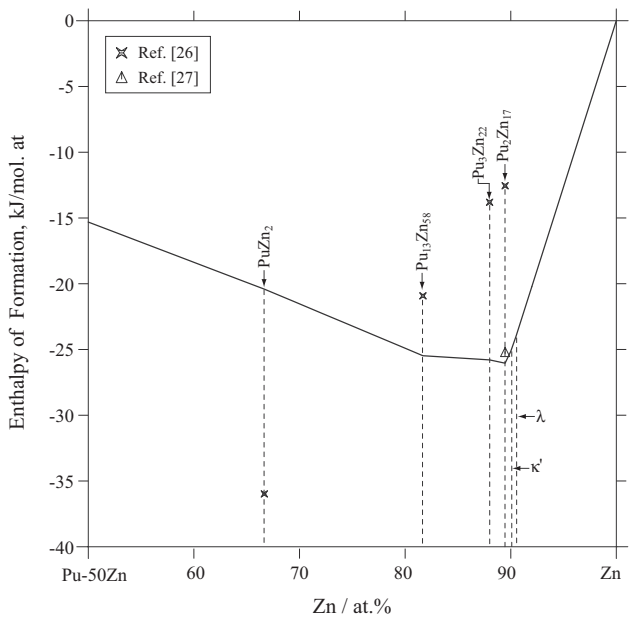

Fig. 4. The calculated enthalpies of formation of intermediate phases with the reference states of (αPu) and (Zn) at 25 °C in the Pu–Zn system compared with the experimental data [26,27].

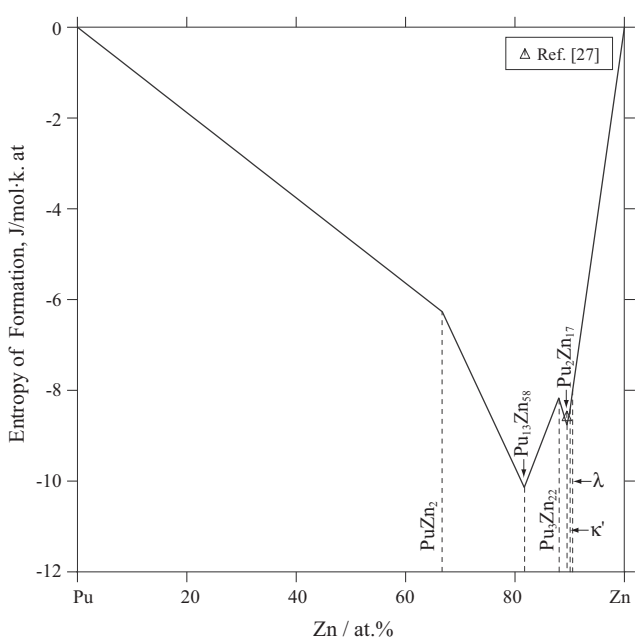

Fig. 5. The calculated entropies of formation of intermediate phases with the reference states of (αPu) and (Zn) at 207°Cin the Pu–Zn system compared with the experimental data [27].

4. Optimized results and discussion

The thermodynamic parameters of each phase were optimized in the THERMO-CALC [34] software, the experimental data were used as input to the program. Each piece of selected information was given a certain weight by the importance of data, shown in many aspects, e.g. errors in experiments, precision of equipment and update of knowledge et al. Apart from the uncertainties in original publications, the authors' judgments by examining all experimental data simultaneously also played a vital role. In the present work, almost all the above experimental information was used in the optimization process. All the optimized thermodynamic parameters of each phase in the Pu–Zn and La–Pu systems are listed in Tables 2 and 3, respectively.

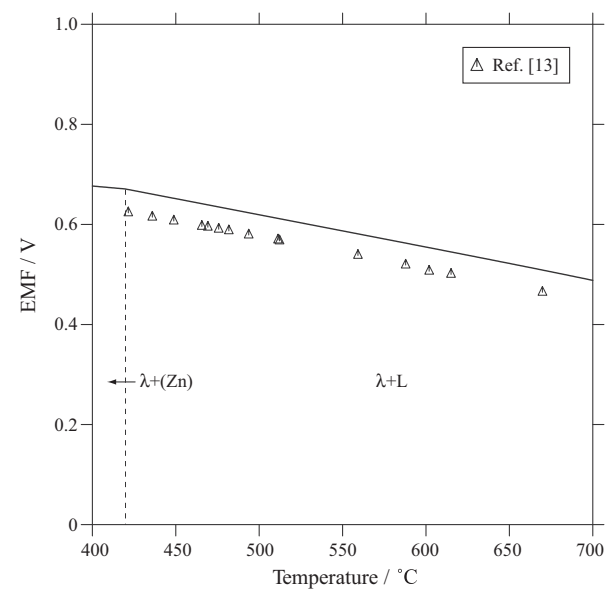

Fig. 6. The calculated EMF of galvanic cell with reference states of (αPu) and (Zn) in Pu-97.8Zn (at.%) alloy compared with the experimental data [13].

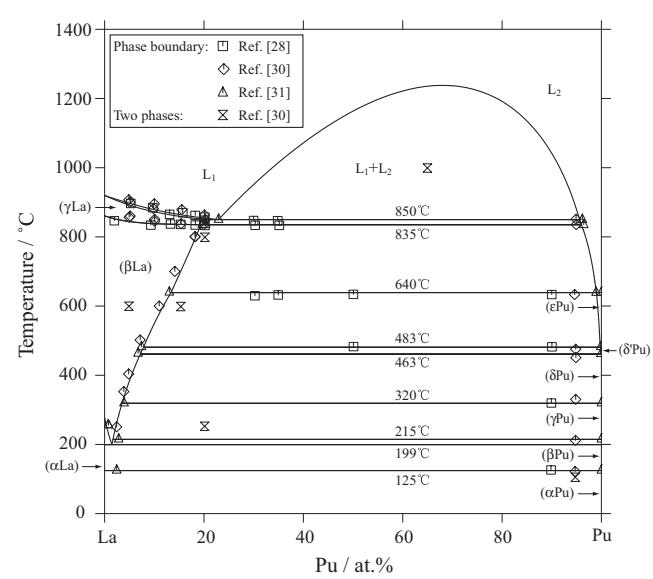

Fig. 7. The calculated phase diagram of the La–Pu system compared with the experimental data [28,30,31].

4.1. The Pu–Zn system

The calculated Pu–Zn phase diagram compared with the experimental data [13,17,18,20,21,25] is shown in Fig. 3a, and the detailed phase diagram in the Zn-rich portion is presented in Fig. 3b. The calculated phase diagram is consistent with the phase diagram reviewed by Massalski et al. [25]. However, the Zn composition of (εPu) in the $PuZn_2$ + (εPu) ↔ (δPu) reaction, the temperature of the L + Pu_3Zn_{22} ↔ $Pu_{13}Zn_{58}$ reaction and the calculated liquidus curve near the Pu-rich side are slightly different from Cramer's results [18,21]. The temperature of the peritectoid reaction, (εPu) + (δPu) ↔ (δ′Pu), is predicted to be 483 °C in this work. All invariant reactions and special points in the system compared with experimental data [18,20,21,25] are summarized in Table 4.

As for the thermodynamic properties, the calculated enthalpies and entropies of formation at 50 °C in contrast with the experimental values [26,27,35] are shown in Figs. 4 and 5, respectively. The calculated enthalpy and entropy of formation of the Pu_2Zn_{17} phase are in good agreement with the calculated values by Chiotti et al. [27] extrapolated from EMF (electromotive forces) measurements of Johnson et al. [13]. However, the calculated enthalpies are different from the values of Miedema's [26]. For the data from Miedema's work [26] are only estimated results, this result is acceptable.

The EMF data of the Pu–97.8Zn (at.%) alloy was calculated through the following formula:

$$E = -\frac{\mu}{ZF} \tag{7}$$

where μ is the chemical potential of Pu in liquid phase, F is the Faraday constant and Z is the transferred electron number in the galvanic cell, Pu(s)|$PuCl_3$, KCl–LiCl (eutectic)|Pu–Zn (two-phase alloy) [13]. A similar tendency of the EMF of galvanic cell in Pu–97.8Zn (at.%) alloy compared with the experimental data [13] is obtained, as shown in Fig. 6.

4.2. The La–Pu system

The calculated La–Pu phase diagram compared with the experimental data is shown in Fig. 7 [28,30,31]. As shown in Table 5, the calculated phase diagram has reached a reasonable agreement with the reviewed phase diagram by Okamoto et al. [31]. The invariant reactions nearly located at the same temperatures and compositions as reported in the involved experiments [28,30] with the maximum deviations of 2 °C and 0.7 at.% Pu. There is some acceptable difference on the shape of the liquid miscibility gap between the calculated diagram and reviewed one [31], and experimental data is available on the limits of the liquid miscibility gap. A new eutectoid transforming, (βLa) ↔ (αLa) + (βPu), which has not been mentioned in all the literature data, is predicted to take place at 199 °C in this work, and this could be used to assist with further investigation. All invariant reactions and critical points in the La–Pu system compared with experimental data [28,30,31] are

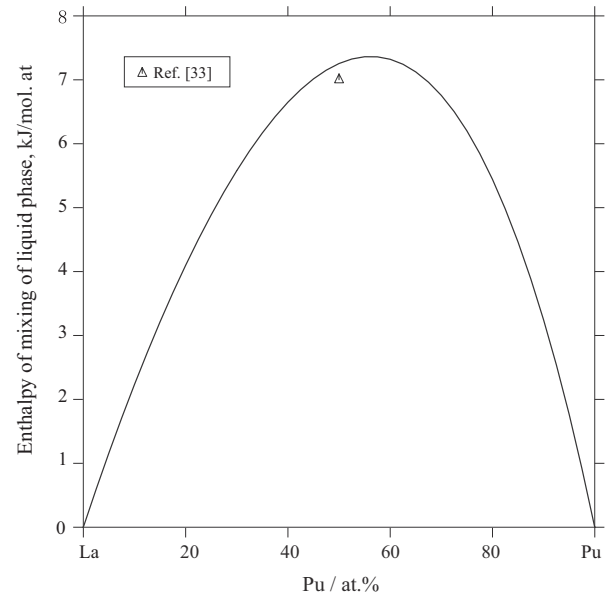

Fig. 8. The calculated enthalpies of mixing of liquid phase in the La–Pu system at 1800 °C with reference states of pure elements of La (liquid) and Pu (liquid) [33].

Table 5
Experimental data and calculated invariant reactions in the La–Pu system [28,30,31].

Reaction type	Reaction	Pu (at.%)			T (°C)	Reference
Monotectic	L_1 ↔ (γLa) + L_2	23	20	–	846	[28]
		23	20	–	850 ± 3	[30]
		23	20	95.7	850 ± 3	[31]
		22.9	20.4	95.6	850	This work
Catatectic	(γLa) ↔ L_2 + (βLa)	18	–	16	833	[28]
		20	–	19	835	[30]
		20	96	19.1	835 ± 2	[31]
		19.7	96	19.3	835	This work
Eutectic	L_2 ↔ (βLa) + (εPu)	–	–	–	632 ± 4	[30]
		98.5	12.8	98.8	~640	[31]
		98.9	13.4	99	639	This work
Eutectoid	(εPu) ↔ (βLa) + (δ′Pu)	–	–	–	475 ± 3	[30]
		99.7	7.1	99.8	~483	[31]
		99.7	7.8	99.8	481	This work
Eutectoid	(δPu) ↔ (βLa) + (γPu)	–	–	–	330	[30]
		100	3.6	100	~320	[31]
		100	3.5	100	319	This work
Eutectoid	(γPu) ↔ (βLa) + (βPu)	100	2.3	100	~215	[31]
		100	1.7	100	215	This work
Eutectoid	(βLa) ↔ (αLa) + (βPu)	1.5	0	100	199	This work
Peritectoid	(βLa) + (δ′Pu) ↔ (δPu)	–	–	–	450 ± 1	[30]
		6.8	99.8	99.8	~463	[31]
		7.2	99.8	99.8	463	This work
Polymorphic	(βPu) ↔ (αPu)	100	100	0	~125	[31]
		100	100	0	125	This work

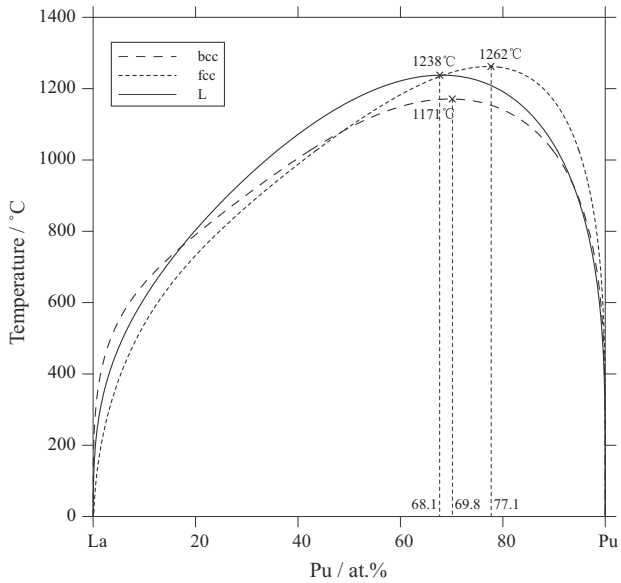

Fig. 9. The calculated stable and metastable miscibility gaps of bcc, fcc, liquid phases in the La–Pu system.

summarized in Table 5. Based on the thermodynamic description evaluated in this work, stable and metastable miscibility gaps of the bcc, fcc and liquid phases are estimated as shown in Fig. 9, which can provide information for further experiment.

In addition, the calculated enthalpy of mixing of liquid phase at 1400 °C agrees well with the Takeuchi's estimation [33], as shown in Fig. 8.

5. Conclusions

The phase diagram and thermodynamic properties in the Pu–Zn and La–Pu binary systems were critically assessed by combining the thermodynamic models with the available literature data. A series of optimized thermodynamic parameters rendering a self-consistent description of the phase relations and thermodynamic properties was derived in the Pu–Zn and La–Pu systems. Phase equilibria and properties calculated with the description derived in the present work could be utilized as a first estimate to better plan such experiments, and could be extrapolated in complex Pu-based systems in the future.

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